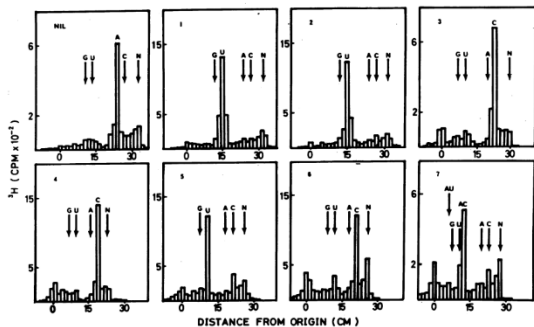


strongly promoted the binding of formylmethionine-tRNA to bacterial ribosomes and functioned as a translational “phaser” that fixed the correct reading frame during initiation. Their experiments with synthetic messenger RNAs showed that AUG was far more effective than other candidate initiation codons such as UUG, providing one of the first mechanistic explanations for how ribosomes recognize translational start sites and establishing AUG as the principal bacterial start codon.

During the late 1960s and early 1970s, studies of prokaryotic translation initiation revealed that ribosomes recognize specific nucleotide sequences upstream of the AUG start codon. At the John Curtin School of Medical Research at the Australian National University, John Shine and Lynn Dalgarno identified a purine-rich sequence upstream of bacterial initiation codons that was complementary to the 3' terminus of 16S ribosomal RNA (Shine and Dalgarno, 1974). Through comparative sequence analysis and ribosome-binding experiments, they demonstrated that base pairing between mRNA and rRNA positioned the ribosome at the translational start site and promoted efficient initiation of protein synthesis.

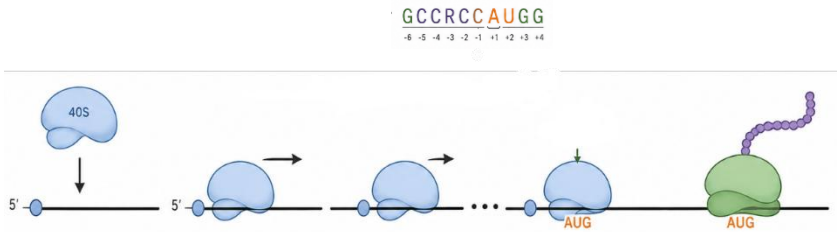
The Shine–Dalgarno model

16S RNA was stepwise degraded, terminally labeled with radioactive [^3H]iNzd, digested with pancreatic ribonuclease, and analyzed by paper electrophoresis using nucleotide standards. After seven degradation cycles, the digestion product was identified as ApC-iNzd, revealing part of the terminal RNA sequence.



The discovery established the Shine–Dalgarno sequence and provided one of the first mechanistic explanations for how ribosomes identify initiation codons in prokaryotes. Because bacterial systems dominated early molecular biology, the Shine–Dalgarno model strongly influenced assumptions about translation initiation in all organisms. However, the absence of comparable motifs in eukaryotic mRNAs, together with the presence of long and structurally complex 5' untranslated regions (5'UTRs), suggested that eukaryotes used fundamentally different mechanisms for start codon recognition. These observations ultimately motivated the work of Marilyn Kozak and others, leading to the ribosomal scanning model and recognition of the 5'UTR as a critical regulatory element in eukaryotic gene expression.

The scanning model of ribosomal initiation



Marilyn Kozak's pioneering studies in molecular biology fundamentally transformed understanding of eukaryotic translation initiation by revealing how ribosomes recognize the correct start codon on messenger RNA. Through systematic mutational analyses in the late 1970s and 1980s, Kozak demonstrated that nucleotide sequences flanking the AUG initiation codon strongly influence translational efficiency, culminating in the definition of the canonical “Kozak sequence” (GCCRCCAUGG), now recognized as a central determinant of translational initiation in eukaryotes (Kozak, 1984; Kozak, 1987). Her work established the scanning model of ribosomal initiation, in which the 40S ribosomal subunit binds near the 5' cap of mRNA and scans downstream to identify an appropriate AUG codon embedded within an optimal sequence context. These discoveries provided a mechanism for understanding gene regulation at the translational level and implications for virology, developmental biology, cancer research, and biotechnology, and ultimately the design of synthetic expression constructs and mRNA therapeutics (Hinnebusch, 2011). Kozak's contributions remain foundational to contemporary RNA biology and continue to shape investigations into translational control and post-transcriptional gene regulation.

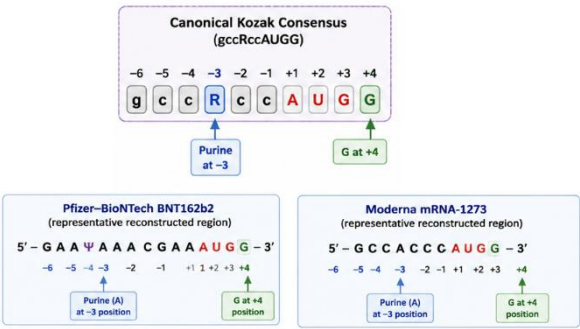
By the late 1980s, strong experimental evidence supported the ribosomal scanning hypothesis. In The scanning model for translation, Kozak and others consolidated findings showing that ribosomes bind near the capped 5' terminus and scan linearly through the 5' UTR until encountering an AUG codon in a favorable sequence context. The model integrated evidence regarding upstream AUGs, RNA secondary structure, and inefficient initiation sites, demonstrating that the 5'UTR functions as a dynamic regulatory region controlling initiation fidelity and ribosome movement.

Functional importance of the 5'UTR transformed our understanding of RNA. Regions once dismissed as non-coding “space” became recognized as central regulatory platforms governing translation initiation, ribosomal fidelity, and protein expression. Kozak’s work established the mechanistic foundations of eukaryotic translation initiation and demonstrated that small variations within non-coding RNA regions can profoundly influence gene expression. These discoveries continue to shape modern biotechnology, synthetic biology, and mRNA therapeutic design, where optimization of the 5' UTR and Kozak sequence remains essential for efficient protein production.

In 1995, Amy E. Pasquinelli, James E. Dahlberg, and Elsebet Lund identified a critical limitation in synthetic mRNA production while studying RNAs generated by bacteriophage RNA polymerases. They demonstrated that conventional cap analogs such as m7GpppG could be incorporated during in vitro transcription in both the correct and reverse orientations, resulting in a substantial proportion of “reverse-capped” transcripts (Pasquinelli, Dahlberg and Lund, 1995). Unlike endogenous eukaryotic mRNAs, whose 5' caps are enzymatically added in a single defined orientation during transcription in the nucleus, synthetic cap dinucleotides can initiate transcription bidirectionally because both guanosine residues possess reactive 3'-OH groups. Reverse-oriented caps bind poorly to the cap-binding protein eIF4E and consequently impair translation efficiency and RNA function. To overcome this problem, Janusz Stepinski and colleagues developed Anti-Reverse Cap Analogs (ARCAs), which contain chemical modifications at the 3' position of the m7G moiety that prevent reverse incorporation while preserving efficient cap recognition (Stepinski et al., 2001). Subsequent studies by Jacek Jemielity and colleagues demonstrated that ARCA-capped mRNAs exhibit substantially enhanced translational efficiency and improved intracellular stability, establishing ARCAs as a foundational advancement in synthetic mRNA technology (Jemielity et al., 2003; Grudzien-Nogalska et al., 2007).

Although ARCAs marked a major advance in synthetic mRNA engineering, newer capping methods such as CleanCap more closely mimic endogenous mammalian mRNA cap structures. ARCAs use chemically modified dinucleotide cap analogs to prevent reverse incorporation during transcription, whereas CleanCap uses a trinucleotide-based co-transcriptional capping approach that ensures correct cap orientation without anti-reverse chemical modifications. ARCAs mainly generate Cap 0 structures, while CleanCap efficiently produces Cap 1 mRNAs with 2'-O-methylation on the first transcribed nucleotide. While ARCAs laid the conceptual and technical foundation for modern synthetic mRNA capping,

CleanCap and related next-generation systems are now widely used in therapeutic mRNA applications.

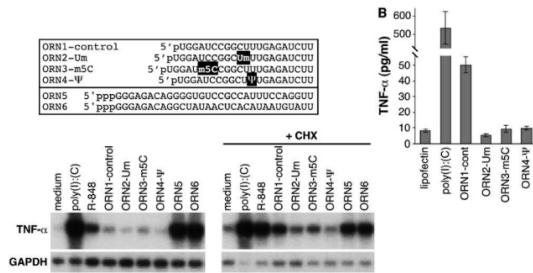


The scanning model established that efficient eukaryotic translation initiation depends on ribosome recruitment at the capped 5' terminus and subsequent scanning of the mRNA by the 40S ribosomal subunit to locate the start codon. This mechanistic framework became essential for modern synthetic mRNA engineering because it revealed that both cap structure and 5' UTR architecture directly determine translational efficiency. Modern mRNA vaccines and protein replacement therapies incorporate optimized UTRs to enabling robust protein expression for therapeutic benefit.

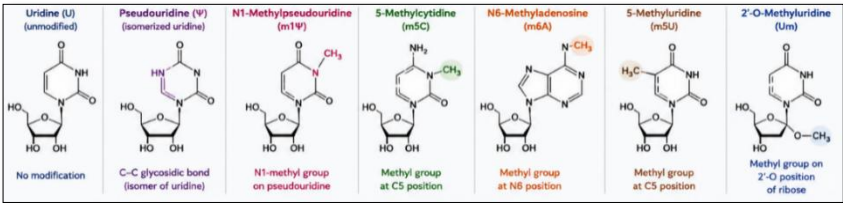
Kozak's discovery that nucleotide context surrounding the AUG codon strongly affects initiation efficiency also directly shaped therapeutic RNA sequence engineering. Nearly all modern therapeutic mRNAs incorporate an optimized Kozak consensus sequence surrounding the start codon to maximize translational output. In mRNA vaccines, for example, inclusion of a strong Kozak context substantially increases antigen production and immune potency. This principle is now routine in mRNA construct design for vaccines, CAR-T engineering, genome editing systems, and protein replacement therapies.

While optimization of the Kozak sequence and 5' UTR architecture greatly improved translation initiation, researchers recognized that efficient therapeutic mRNA expression also depends on the properties of the coding sequence itself. This realization shifted mRNA engineering beyond initiation-site optimization toward broader control of transcript stability, ribosomal elongation kinetics, and innate immune recognition. Modern therapeutic mRNAs therefore incorporate extensive coding-sequence engineering, including codon optimization, modulation of GC content, reduction of inhibitory secondary structures, and incorporation of modified nucleotides such as pseudouridine and N1-methylpseudouridine. These

nucleotide modifications reduce activation of innate immune sensors such as Toll-like receptors and PKR while simultaneously enhancing ribosome processivity, transcript stability, and protein production. Together with optimized cap structures, Kozak contexts, and engineered untranslated regions, these advances transformed synthetic mRNA from an unstable experimental molecule into a clinically viable therapeutic platform.



In 2005, Katalin Karikó, Michael Buckstein, Houping Ni, and Drew Weissman at the University of Pennsylvania identified that synthetic RNA strongly activates innate immune pathways. They showed that unmodified RNA stimulates human Toll-like receptors TLR3, TLR7, and TLR8, triggering dendritic-cell activation and inflammatory cytokine production. Crucially, they demonstrated that naturally occurring nucleoside modifications, including pseudouridine, 5-methylcytidine, N6-methyladenosine, 5-methyluridine, and 2-thiouridine, suppress this immune recognition. This work explained why mammalian RNA, which is rich in modified nucleosides, is less immunostimulatory than bacterial or in vitro-transcribed RNA, and it established pseudouridine modification as a foundational strategy for making synthetic mRNA safer, more stable, and more suitable for therapeutic use (Karikó et al., 2005).



Subsequent work by Brian R. Anderson, Hisaaki Muramatsu, Katalin Karikó, Drew Weissman, and colleagues substantially expanded the mechanistic understanding of how modified nucleosides improve

therapeutic mRNA performance. Building on earlier findings that pseudouridine suppresses Toll-like receptor activation, the group demonstrated that modified nucleosides also inhibit intracellular antiviral defense pathways mediated by interferon-induced 2'–5' oligoadenylate synthetase (OAS) and RNase L (Anderson *et al.*, 2011). Under normal antiviral conditions, unmodified foreign RNA activates OAS enzymes, which synthesize short 2'–5'-linked oligoadenylates that subsequently activate the latent endoribonuclease RNase L. Activated RNase L rapidly degrades intracellular RNA, suppresses protein synthesis, and promotes apoptosis as part of the innate antiviral response. Anderson and colleagues showed that incorporation of modified nucleosides such as pseudouridine markedly reduces activation of OAS and downstream RNase L-mediated RNA cleavage, thereby protecting synthetic mRNAs from intracellular degradation (Anderson *et al.*, 2011). These findings revealed that nucleoside modification does not just reduce sensing through TLR pathways but also suppresses cytoplasmic antiviral surveillance systems that normally limit translation of exogenous RNA.

These advances were crucial to the development of clinically useful mRNA therapeutics. Nucleoside modification greatly improved the stability, translational efficiency, and tolerability of synthetic mRNA platforms. Building on the success of pseudouridine, the field progressed to next-generation modified nucleosides such as N1-methylpseudouridine (m1Ψ), which further increased protein expression while reducing inflammatory signaling and innate immune activation. Both the Pfizer–BioNTech BNT162b2 and Moderna mRNA-1273 SARS-CoV-2 vaccines incorporate m1Ψ in place of uridine throughout the coding region. This modification suppresses activation of Toll-like receptors, PKR, and the OAS/RNase L antiviral pathway while also improving ribosome processivity and intracellular mRNA stability, enabling the high levels of spike protein expression required for strong vaccine-induced immunity. Together, these innovations transformed synthetic mRNA from a highly immunostimulatory experimental molecule into the foundation of modern mRNA vaccines and therapeutic RNA technologies.

A-to-I RNA editing oligonucleotide (ADAR)



CRYO-EM STRUCTURE OF HUMAN ADAR1 IN COMPLEX WITH DSRNA DERIVED FROM HT2C GENE
HOMO SAPIENS
2025-03-26
AUTHORS: DENG, X., GAO, Y.

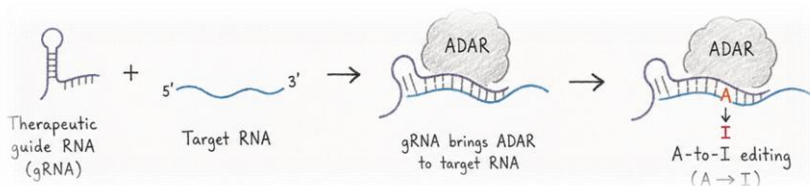
Early work on RNA processing had already shown that cells could alter RNA after transcription, raising the possibility that genetic information did not always pass unchanged from DNA to mature RNA. Researchers studying RNA structure then focused on double-stranded RNA since base-paired RNA molecules form stable duplexes that can influence translation, viral defense and RNA maturation. Against this background, Bass and Weintraub examined *Xenopus laevis* (frog) oocytes and embryos and identified an enzymatic activity that acted specifically on RNA duplexes. They initially described the activity as one that “unwound” double-stranded RNA because treated duplexes lost the behavior expected of fully base-paired RNA (Bass and Weintraub, 1987; Bass and Weintraub, 1988).

Subsequent biochemical work, undertaken to explain why dsRNA duplexes lost their normal base-paired behavior after treatment with the “unwinding” activity, showed that this apparent unwinding reflected chemical modification rather than conventional helicase activity. At The Wistar Institute and the University of Pennsylvania in Philadelphia, Wagner, Smith, Cooperman and Nishikura demonstrated that the activity introduced structural changes into double-stranded RNA through adenosine-to-inosine conversion in mammalian cells and *Xenopus* eggs (Wagner et al., 1989). Polson and colleagues at the University of Utah in Salt Lake City then confirmed the mechanism directly by mass spectrometry, showing that the double-stranded RNA unwinding/modifying activity converted adenosine to inosine (Polson et al., 1991). These findings linked the original dsRNA

observation to RNA editing and led to the definition of ADARs as adenosine deaminases acting on RNA.

Work on glutamate receptor pre-mRNAs then showed how endogenous transcripts create the dsRNA structures required for ADAR activity. Higuchi and colleagues showed that editing at the GluR-B/GluA2 Q/R site required base pairing between an exonic sequence and a downstream intronic editing-site complementary sequence, demonstrating that cellular pre-mRNAs can fold into the local duplexes needed to direct site-specific editing (Higuchi et al., 1993). ADAR1 and ADAR2 were subsequently cloned and characterized as RNA-editing enzymes (Kim et al., 1994; Melcher et al., 1996). ADAR2-mediated editing of the GluA2 Q/R site changes a genomically encoded glutamine codon into an arginine-read codon and limits calcium permeability through AMPA receptors, making this site a central example of physiologically important finding (Higuchi et al., 1993; Wright et al., 2023).

Genome-wide analyses then showed that ADAR activity extends far beyond a small set of coding sites. Levanon and colleagues mapped thousands of A-to-I editing sites in the human transcriptome and showed that most occur in non-coding regions, especially within Alu repeat-derived double-stranded RNA structures (Levanon et al., 2004). Thus, the field shifted from viewing ADARs primarily as enzymes that diversify proteins to recognizing them as broader regulators of non-coding RNA structure, repetitive-element-derived dsRNA and endogenous RNA surveillance.



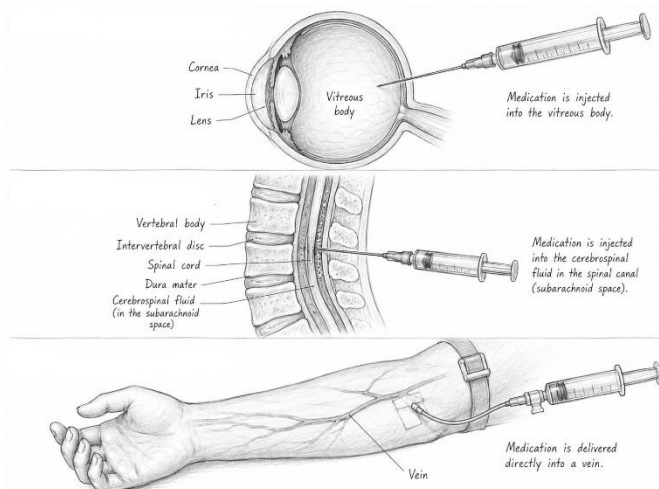
ADAR Rx

Industry efforts aim to redirect ADAR activity with engineered guide RNAs or oligonucleotides that create a programmable RNA duplex around a disease-relevant adenosine, allowing endogenous or engineered ADAR enzymes to install a targeted A-to-I change without permanently altering DNA. Wave Life Sciences has moved this concept into the clinic with WVE-006 for alpha-1 antitrypsin deficiency, reporting the first therapeutic RNA editing in humans in 2024 and subsequent phase 1b/2a data showing therapeutically relevant AAT levels across multiple dose cohorts. ADARx Pharmaceuticals is also pursuing an RNA editing medicines pipeline across

hepatic and extrahepatic diseases, including complement-mediated, cardiovascular, cardiometabolic and central nervous system indications.

While this technology has promise for context-relevant genetic therapeutics, it still faces many of the same practical barriers that have limited other genetic medicine platforms. Chief among these is delivery: getting the therapeutic payload to the right tissue, into the right cell type, at sufficient levels, and with enough durability to produce a meaningful clinical effect. This remains especially challenging for tissues that are difficult to access, such as the central nervous system, skeletal muscle, heart, and certain solid tumors. Even when systemic delivery is possible, biodistribution can be uneven, with uptake concentrated in organs such as the liver or spleen rather than the intended disease-relevant tissue.

Pharmacokinetics and pharmacodynamics of RNA Therapeutics



Pharmacokinetics and pharmacodynamics are central to the development of nucleic acid therapeutics because these modalities do not behave like conventional small molecules or typical biologics. For antisense oligonucleotides, siRNAs, lipid nanoparticle-formulated RNAs and related genetic medicines, the relationship between administered dose, systemic exposure, tissue distribution, intracellular trafficking, target engagement and biological response is often indirect, delayed and highly tissue-dependent. In many cases, plasma concentration is only a partial measure of pharmacologically relevant exposure; the more important question is

whether sufficient active drug reaches the correct tissue, the correct cell type and the correct intracellular compartment.

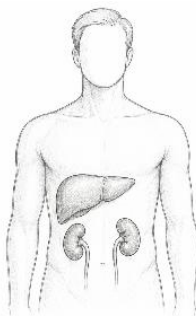
Antisense oligonucleotides are generally single-stranded, chemically modified nucleic acids designed to bind complementary RNA through sequence-specific hybridization. Depending on their design, ASOs may trigger RNase H-mediated degradation of target RNA, alter splicing, block translation or modulation of RNA processing. Their pharmacokinetics are strongly influenced by chemistry modifications, plasma protein binding, nuclease stability and tissue uptake. Phosphorothioate backbone modifications, for example, increase plasma protein binding and reduce rapid renal clearance, supporting broader tissue distribution and longer tissue residence (Geary et al., 2015). ASOs often distribute rapidly from plasma into tissues, followed by slow tissue elimination and prolonged pharmacodynamic activity, meaning that plasma PK alone can substantially underestimate the duration and relevance of tissue exposure (Geary et al., 2015; Burel et al., 2024). In practice, ASO PK/PD therefore requires attention to tissue concentration, intracellular availability, target RNA knockdown and the turnover kinetics of the downstream protein or pathway.

siRNAs are double-stranded RNA molecules that act through the RNA-induced silencing complex, or RISC, to degrade complementary mRNA. Their pharmacology depends on tissue distribution, cell uptake, endosomal trafficking, cytosolic release, guide-strand loading into RISC, target mRNA cleavage and downstream protein reduction. For GalNAc-conjugated siRNAs, hepatocyte delivery is mediated by binding to the asialoglycoprotein receptor, creating a highly targeted liver-delivery platform with distinctive PK/PD behavior (Ayyar and Song, 2024). Mechanistic models of GalNAc-siRNA disposition incorporate receptor-mediated uptake, intracellular sequestration, slow degradation and RISC-dependent mRNA silencing, reflecting the fact that the pharmacologically active species is not simply circulating siRNA but intracellular, functionally loaded guide strand (Ayyar and Song, 2024). This helps explain why siRNA pharmacodynamic effects can last weeks to months even after plasma concentrations have fallen.

A key pharmacology feature of these modalities is durability. Studies of GalNAc-siRNA conjugates suggest that chemically stabilized siRNA can persist in acidic intracellular compartments and be slowly released over time, enabling continued RISC loading and prolonged gene silencing (Brown et al., 2020). This creates an important PK/PD distinction between systemic exposure, tissue exposure and productive intracellular exposure. Two siRNAs may show similar early knockdown but differ markedly in duration depending on chemical stability, trafficking route and intracellular depot behavior. Delivery route matters: GalNAc conjugates and

LNP-formulated siRNAs can produce different onset and duration profiles, consistent with differences in intracellular trafficking and cytosolic release kinetics. A related durability principle is seen with Spinraza® (nusinersen), an intrathecally administered antisense oligonucleotide for spinal muscular atrophy. Although nusinersen acts through splice modulation rather than RNAi, its prolonged clinical dosing interval illustrates how oligonucleotide chemistry, tissue retention and intracellular persistence can uncouple short systemic exposure from sustained pharmacodynamic effect. Following loading doses, nusinersen is maintained with dosing every four months, supported by slow elimination from cerebrospinal fluid and distribution within central nervous system tissues (FDA, 2026). Its reported CSF terminal half-life of approximately 135–177 days further highlights that the durability of oligonucleotide therapeutics depends less on plasma PK alone and more on residence time in the relevant tissue compartment and continued productive engagement of intracellular RNA-processing machinery (FDA, 2026; Chiriboga et al., 2016).

Lipid nanoparticles add to another level of complexity because both the RNA payload and the delivery vehicle contribute to pharmacology, biodistribution and safety. LNPs are used to deliver mRNA, siRNA, gene-editing systems and other nucleic acid payloads by protecting RNA from degradation, promoting cellular uptake and enabling intracellular release. Their behavior is shaped by particle size, ionizable lipid chemistry, PEG-lipid content, cholesterol and helper lipid composition, surface charge, protein corona formation and route of administration. For LNP-formulated RNA therapeutics, PK/PD therefore includes not only where the nanoparticle goes, but also whether the payload escapes the endosome and becomes biologically active. For mRNA LNPs, the primary pharmacodynamic outcome is protein expression rather than target knockdown. The PK/PD sequence therefore links dose to LNP biodistribution, cellular uptake, endosomal escape, cytosolic mRNA availability, translation, protein expression, protein turnover, and the resulting biological effect.



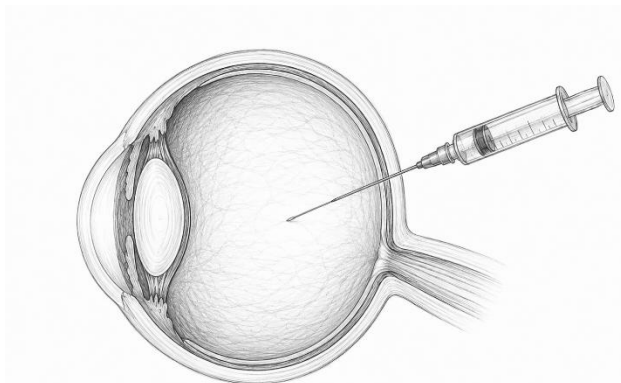
An example of how formulation can reshape oligonucleotide PK/PD is provided by Yu et al. (1999) at Isis pharmaceuticals (no called Ionis Pharmaceuticals), who compared a phosphorothioate antisense oligonucleotide against Ha-Ras delivered either as a non-liposomal saline formulation or encapsulated in long-circulating PEGylated “stealth” liposomes. The study showed that liposomal encapsulation prolonged systemic exposure: plasma half-life increased from approximately 1.1 hours for the saline formulation to approximately 57.8 hours for the PEGylated liposome formulation, while clearance fell from about 54.0 to 0.892 ml/hr/kg. The liposomal formulation also protected the oligonucleotide from nuclease-mediated metabolism, with mostly intact oligonucleotide compound detected in plasma and blood over time, whereas the saline formulation showed rapid disappearance and more chain-shortened metabolites. Tissue distribution was also altered: non-liposomal oligonucleotide showed high renal exposure, consistent with uptake and clearance by kidney pathways, while PEGylated liposomes shifted exposure toward reticuloendothelial-rich tissues such as liver, spleen and lymph nodes, with slower tissue uptake and longer residence. This illustrates a broader PK/PD principle for nucleic acid therapeutics: formulation can change not only circulation time, but also the dominant clearance route, metabolic stability, tissue exposure pattern and potential therapeutic index. In the context of oncology, the authors highlighted that long-circulating PEGylated liposomes may exploit leaky tumor vasculature and sites of inflammation, supporting improved delivery to tissues with increased vascular permeability (Yu et al., 1999).

PK/PD	Non-liposomal oligonucleotide	PEGylated stealth liposome
Plasma half-life	1.1 hours	~57.8 hours
Clearance	54.0 ml/hr/kg	0.892 ml/hr/kg
Metabolism	Rapid degradation; more shortened metabolites	Greater protection; more intact oligonucleotide
Distribution pattern	High kidney exposure	More liver, spleen and lymph-node exposure
Translational implication	Fast clearance, renal uptake, short exposure	Longer circulation, slower clearance, possible enhanced delivery to permeable tissues

Clinical compounds with route of administration

Compound	Base	Backbone	Sugar	Conjugation	Route
Fomivirsen	21-mer antisense DNA	Full phosphorothioate	DNA	None	Intravitreal injection
Mipomersen	Gapmer ASO	Phosphorothioate	2'-MOE + DNA gap	None	Subcutaneous injection
Eteplirsen	30-mer antisense	PMO	Morpholino	None	Intravenous infusion
Nusinersen	18-mer	Phosphorothioate	2'-MOE	None	Intrathecal injection
Inotersen	Gapmer ASO	Phosphorothioate	2'-MOE	None	Subcutaneous injection
Golodirsen	Antisense	PMO	Morpholino	None	Intravenous infusion
Volanesorsen	Gapmer ASO	Phosphorothioate	2'-MOE	None	Subcutaneous injection
Viltolarsen	Antisense	PMO	Morpholino	None	Intravenous infusion
Casimersen	Antisense	PMO	Morpholino	None	Intravenous infusion
Tofersen	20-mer gapmer	Mixed PS/PO	2'-MOE	None	Intrathecal injection
Eplontersen	Gapmer ASO	Mixed PS/PO	2'-MOE	GalNAc	Subcutaneous injection
Pegaptanib	RNA aptamer	Modified RNA	Modified ribose	PEG	Intravitreal injection
Avacincaptad pegol	RNA aptamer	Modified RNA	Modified ribose	PEG	Intravitreal injection
Patisiran	siRNA	Phosphodiester	2'-OMe	LNP	Intravenous infusion
Givosiran	siRNA	PS modified	2'-F/2'-OMe	GalNAc	Subcutaneous

Lumasiran	siRNA	PS modified	2'-F/2'-OMe	GalNAc	injection Subcutaneous injection
Inclisiran	siRNA	PS modified	2'-F/2'-OMe	GalNAc	Subcutaneous injection
Vutrisiran	siRNA	PS enhanced	2'-F/2'-OMe	GalNAc	Subcutaneous injection
Nedosiran	siRNA	PS/PO	2'-F/2'-OMe	GalNAc	Subcutaneous injection
Miravirsen	Anti-miR	Phosphorothioate	LNA/DNA	None	Subcutaneous injection
RG-101	Anti-miR	Modified ASO	Not specified	GalNAc	Subcutaneous injection



Pharmacokinetic studies showed that Fomivirsen distributes primarily within ocular tissues following intravitreal injection and had sustained concentrations with a vitreous half-life of approximately 55 hours in humans and an estimated retinal half-life of up to 78 hours in animal models (Geary, Henry and Grillone, 2002). Metabolism occurs via endo- and exonuclease-mediated degradation into shorter oligonucleotide fragments, with minimal systemic exposure due to localized administration.

For phosphorothioate (PS) ASOs, which are typically ≥ 85 –99% protein bound, renal clearance of intact drug is minimal, with studies showing that < 1 –4% of the administered dose is recovered in urine within 24 hours, reflecting predominant elimination via slow nuclease-mediated metabolism in tissues (Geary et al., 2024; mipomersen data). In contrast, low protein-binding chemistries such as phosphorodiamidate morpholino oligomers (PMOs) undergo substantial renal elimination, with a large fraction of the administered dose excreted unchanged in urine due to limited plasma protein binding and rapid glomerular filtration (Shadida et al., 2021). siRNA shows an intermediate and modality-dependent profile: unmodified siRNA is rapidly cleared by renal filtration, whereas clinically used GalNAc-conjugated siRNAs exhibit reduced renal clearance, with approximately 12–25% of the dose recovered unchanged in urine and the remainder undergoing tissue uptake and intracellular metabolism (Habtemariam et al., 2021; vutrisiran data).

Nucleases are fundamental to life. These enzymes catalyze the cleavage of DNA and RNA, enabling essential processes such as DNA replication (e.g., removal of RNA primers and proofreading), DNA repair